ELSEVIER

Tectonophysics 332 (2001) 69–92

TECTONOPHYSICS

www.elsevier.com/locate/tecto

The rotations of the Variscides during the Carboniferous collision: paleomagnetic constraints from the Vosges and the Massif Central (France)

J.B. Edel*

Institut de Physique du Globe de Strasbourg (CNRS-ULP), 5 rue Descartes, 67084 Strasbourg Cedex, France

Abstract

This study aims at completing and refining results from previous paleomagnetic investigations carried out in various outcrops of the European Variscan belt. Sampling was performed in Visean effusive and intrusive rocks of the southern Vosges basin and the northern Massif Central, all emplaced in the time range 345–340 Ma. Stepwise thermal demagnetization of 254 specimens from 13 sites of the Doller Valley and nine sites of the Massif Central has yielded a very similar paleomagnetic record for both areas. With the exception of post-Carboniferous overprints isolated in the southern Vosges, two groups of directions, with both normal and reversed polarity have been evidenced. The first group of directions, labelled *C*, occurs in both areas and is characterized by scarce pre-tectonic magnetizations (two sites) and by syn-tectonic overprints (five sites) acquired during the Late Visean convergence, i.e. in the time-range 340–328 Ma. The mean pole C_o is 28°S/45°E ($k = 139$, $\alpha_{95} = 5°$). The main group, labelled *B*, shows a negative fold test and a positive reversal test, indicating that the magnetizations are post-tectonic overprints, acquired after the main tightening phase, i.e. after 328 Ma. At a few sites in both areas were evidenced syn-tectonic *B* magnetizations that are in favour of an overprinting in the context of post-Visean extension tectonics. The mean post-tectonic *B* direction of the Doller Valley (southern Vosges) 62°/−25° ($k = 52$, $\alpha_{95} = 6°$, $N = 11$; VGP: 8°N/127°E) is consistent with the mean post-tectonic *B* direction 72°/−23° ($k = 126$, $\alpha_{95} = 5°$, $N = 6$; VGP: 3°N/114°E) obtained from the tuffs, the grey microgranite and the dyke intrusive into the microgranite of the Massif Central. The occurrence of similar paleomagnetic records in the eastern and central Variscides implies that, from the Massif Central to the Sudetes, the belt has probably moved as one block, since Early Carboniferous.

The change in inclination of the *C* components suggests a southward drift of the Moldanubian and Saxothuringien Variscides in the Late Visean (340–328 Ma). According to geological features, in the same time, the Rhenohercynian crust was underplating the Saxothuringian arc and fore-arc zone. Subsequent to 328 Ma, the drift turned into a rapid clockwise rotation by about 70° around a pole located on southeastern corner of Baltica, leading to the closure of the Rhenohercynian oceanic basin. In the Namurian, large-scale overprinting (*B* components), likely related to post-rotation extension, affected the internal belt. The convergence continuing, the Variscides underwent a second clockwise rotation by about 45° in the Late Namurian–Westphalian (320–305 Ma). Stacking of the Rhenohercynian sedimentary cover and thrusting over the London–Brabant basement may be related with this motion. © 2001 Elsevier Science B.V. All rights reserved.

Keywords: paleomagnetism; magnetic overprinting; tectonic rotations; Carboniferous; Vosges; Massif central; Variscides

1. Introduction

In the last 20 years, several paleomagnetic investigations have been carried out in the Massif Central,

* Tel.: +33-88416300; fax: +33-88616747.
E-mail address: jbe@eost.u-strasbg.fr (J.B. Edel).

0040-1951/01/$ - see front matter © 2001 Elsevier Science B.V. All rights reserved.
PII: S0040-1951(00)00250-X

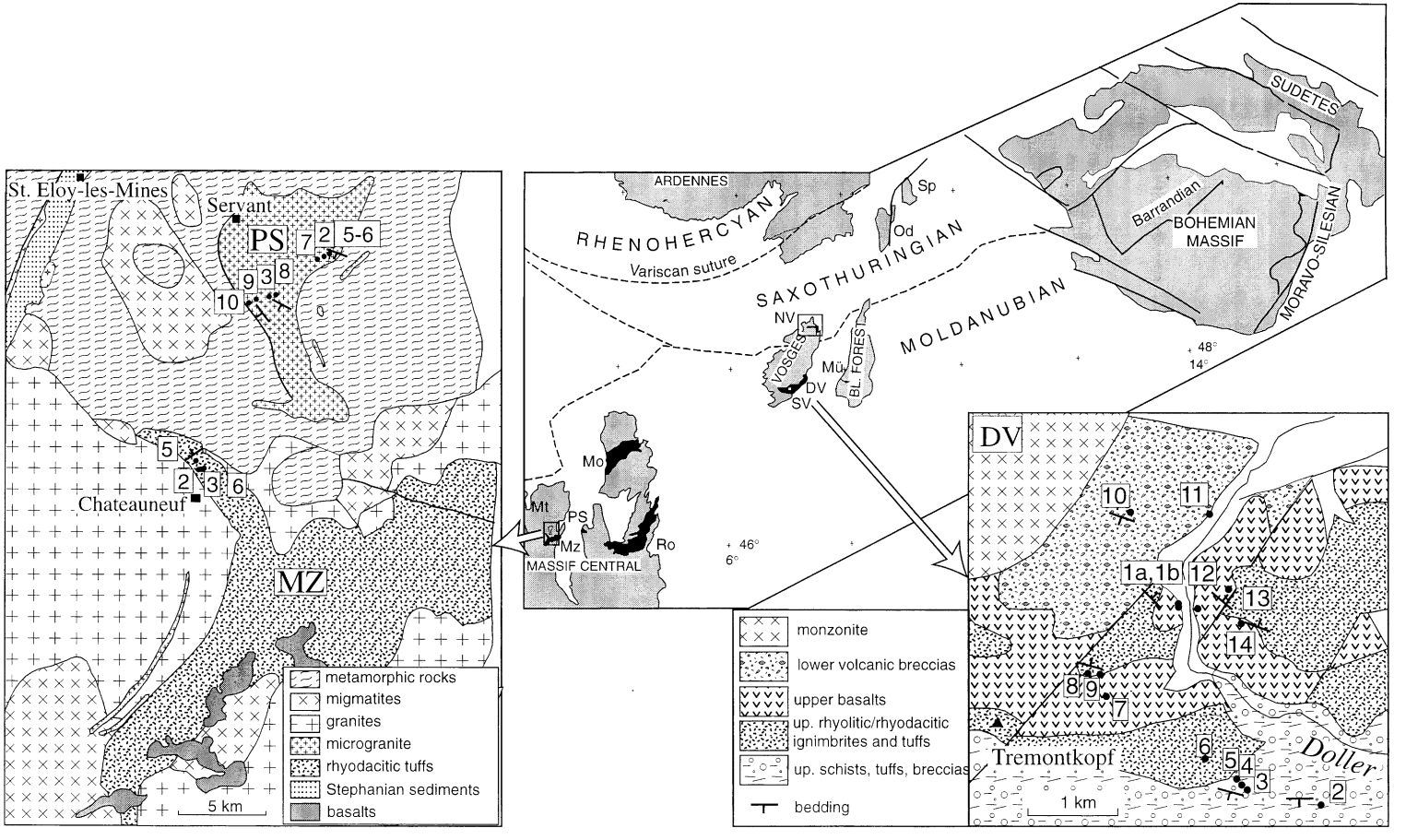


Fig. 1. Geological maps, location and bedding of the paleomagnetic sites. Sp: Spessart, Od: Odenward, NV: Northern Vosges, SV: Southern Vosges, DV: Doller Valley, Mü: Münstertal ignimbrites, Mo: Morvan, Ro: Roannais, Mz: Manzat tuffs, PS: Pouzol-Servant microgranite, Mt: Montmarault granite.

the Vosges, the Black Forest, the Odenwald and the Spessart (Fig. 1) (Edel, 1987a,b; Edel et al., 1981, 1986; Edel and Wickert, 1991; Edel and Lewandowski, 1993), in order to detect possible block rotations and latitudinal drifts in relation with the Carboniferous convergence of Gondwana and Laurussia (Matte, 1991). The investigated units, which consist of Visean volcanic and plutonic rocks of the Massif Central, Vosges, Odenwald and Spessart, and of metamorphic rocks of the Massif Central and Spessart, are in general magnetically overprinted. Due to various rock types, various locations with respect to faults, various magnetic minerals and grain sizes, overprinting was not synchronous, and the respective paleomagnetic records cover a long period ranging from the Late Visean up to the Jurassic (Edel et al., 1997, 1998). The post-Carboniferous overprints, the poles of which fit the Permian to Jurassic part of the European APWP, have been removed (Van der Voo, 1990; Edel and Schneider, 1995; Edel and Duringer, 1997). Plotting the remaining directions of a same massif on a sphere or on a cylindrical projection shows eight clusters consisting of four groups with antipodal directions, labelled $A1$, B, C_p and C_n (Edel and Lewandowski, 1993). The $A1$ components are characterized by SSW declinations and shallow positive inclinations (or opposite). They show a negative fold test indicating a magnetic overprinting after the main folding phase. As the corresponding VGPs fairly fit the mean Late Carboniferous pole of Variscan Europe (Edel and Schneider, 1995), the same age was assigned to the overprints. The present paper focuses on the B and C directions in the central part of the Variscides that were interpreted as the oldest magnetizations. In the light of recent and older radiometric ages as well as of fold tests, an attempt to date the magnetizations is made, in order to interpret the directions in terms of geodynamics of the Variscides in Carboniferous times.

2. Geological setting and sampling

2.1. The Doller Valley Visean volcanites

In the southern Vosges, the investigation was limited to a small area of the Early Carboniferous basin, where the Middle Visean layers of volcanites, volcanic breccias and schists are dipping either towards north or south (Fig. 1). The area consists of a thick sequence of Late Devonian to Late Visean sedimentary, and volcanic units intruded, in the Late Visean, by granitic plutons associated with the volcanism (Fig. 1). The Late Devonian to Middle Visean volcanites show a tholeitic affinity that testifies to the opening of the basin during an extensional period (Coulon et al., 1979; Schneider, 1990, 1994; Schneider et al., 1989). In the Late Visean, the volcanism evolved from calc-alkaline to shoshonitic affinities. At the end of the volcanic and sedimentary activity, in the latest Visean, a major deformation, resulting from the S–N tightening of the Variscan belt, affected the basin.

The investigated series belongs to the top of the lower unit that was emplaced in the Middle Visean under marine conditions. From the base to the top, sampling was performed in: lower breccias, composed of elements of pyroclastic and basaltic rocks sealed in a rhyolitic matrix (sites DV10, 11); upper basaltic layer (DV1, 1b, 7, 9, 12, 13, 14); upper rhyolitic and rhyodacitic ignimbrites and tuffs (DV6, 8); and uppermost tuffs and breccias (DV2, 3, 4, 5) interbedded with marine sediments. A U–Pb age of 345 ± 2 Ma dates a low-K rhyolite located between DV6 and the Tremontkopf (Schaltegger et al., 1996) (Fig. 1). The monzonites and granite des Ballons intruded the basin a short time later, respectively, at 342 ± 1 and 340 + 4/ − 2 Ma. The calc-alkaline to shoshonitic volcanites of the upper Visean have exhibited the same ages around 340 Ma as the granites (Schaltegger et al., 1996).

In thin sections, abundant chlorite and albitization of plagioclases indicate green-schist metamorphism at all investigated sites.

2.2. The Visean tuffs of Châteauneuf

The dacitic volcanites of Manzat are divided into two units: the lower complex that is composed of pyroclastic breccias, lavas, lappili tuffs and ignimbritic levels; and the upper level that consists of ignimbritic ash-tuffs (Vennat, 1985). These tuffs are similar to the dacitic tuffs of the southern limb of the southern Vosges basin. Old Rb–Sr dating of biotite and whole rock provided ages in the range 335–342 Ma (Vialette, 1962; Roques et al., 1971) very

Table 1
Paleomagnetic results from the upper Doller Valley volcanites (southern Vosges, 47.8°N/6.9°E). T_{ub}: maximum unblocking temperature; n: number of specimens; N: number of sites; D_0/I_0: in situ declination/inclination; D_c/I_c: declination/inclination corrected for dip; k/α_{95}: Fisher statistic parameters; Comp: label of the component; B_{pt}: post-tectonic direction; B_{at}: ante-tectonic direction. N/R: normal/reversed polarity. The directions in bold characters are taken for the calculation of the VGP

Site	Rock type	Bt (°C)	n/N	D (°)	I (°)	D_c	I_c	k	α_{95} (°)	Comp	VGP (°N)	VGP (°E)
DV1a	Basalt	580–600	6	**245**	**23**	239	1	93	7	B_{pt}	7	124
		580–600	4	**217**	**20**	216	−11	41	14	*A*1	23	147
DV2	Tuff (fine gr.)	400–580	4	**250**	**29**	235	2	43	14	B_{pt}	1	122
DV3	Tuff (coarse gr.)	500–580, > 600	6	**62**	**−19**	63	22	38	11	B_{pt} (N + R)	11	125
DV4	Basalt	400–500	4	**76**	**−19**	74	10	36	15	B_{pt}	2	114
DV5	Tuff	500	6	**63**	**−26**	57	−8	40	11	B_{pt}	7	126
DV6	Tuff (fine gr.)	300–580	5	**42**	**−30**	38	−7	31	14	B_{pt}	16	145
DV8	Basalt	580	6	246	−11	**249**	**22**	52	9	B_{at}	5	120
		580–590	7	**224**	**31**	261	67	42	9	B_{pt}	14	144
DV9	Keratophyre	530–550	6	**233**	**21**	258	57	49	10	B_{pt}	15	133
		580–595	6	**57**	**−50**	140	−68	64	8	B_{st}		
		580–595	6	143	4	133	22	38	10	C_{st}	−18	56
DV10	Pyroclastic	530–580	6	308	17	**295**	**24**	36	11	C_{at}	−26	87
		580–590	8	261	53	**237**	**29**	75	6	B_{at}	9	132
		580–600	4	**242**	**23**	238	−3	370	5	B_{pt}	9	126
DV12	Basalt	< 400, > 600	5	**78**	**−22**	85	−37	22	16	B_{pt}	−1	113
DV13	Basalt	530–550	4	**66**	**−28**	73	−45	126	8	B_{pt}	4	125
		580	9	211	33	195	56	36	9	*A*1	19	156
DV14	Basalt	550–580	6	328	4	**326**	**−14**	33	12	C_{at}	−27	46
Mean B_{pt}			11	**62**	**−25**			52	6		**8**	**127**
						63	−17	7	18			
Mean B_{at}			2	252	21			< 1				
						243	**26**	79			**7**	**126**
Mean *B* with normal polarity			6	**62**	**−24**			42	10			
Mean *B* with reversed polarity			7	**240**	**26**			76	7			
All *B* components, ante- and post-tectonic			13	**62**	**−25**			58	5		**8**	**127**
Mean *C*			3	320	6			30	22		−34	58
						313	4	8	45		−28	64
Mean *A*1			2	**214**	**26**			67			**21**	**151**
						208	22	3				

consistent with the recent U–Pb ages of the tuffs and ignimbrites of the southern Vosges. In a previous study, the upper unit has not showed any characteristic magnetizations, while the two sites collected in the pyroclastic breccias of the Châteauneuf area (Mz2, Mz3) exhibited typical *B* directions and scarce *C* directions (Fig. 1). Therefore, sampling was extended to two additional sites of the Châteauneuf section, one below Mz2 and Mz3 (Mz5) and the second on the top of the series (Mz6). Along the investigated section, fiammes, planar structures, flattened inclusions and columnar jointing indicate a north–northwestward bedding.

2.3. The Pouzol-Servant laccolite

The Pouzol-Servant microgranite is a hypovolcanic laccolite, associated with the dacitic volcanism, and intrusive into the metamorphic units of the Sioule (Vennat, 1985) (Fig. 1). Microgranite and volcanites are, respectively, interpreted as the lower and upper levels of a caldera. The root of the laccolite runs along the fault that marks the western border. Sampling was performed along the gorge of the Sioule river, in the generally light grey-coloured granite (PS2, 6, 7, 9) and occasionally pink or reddish granite (PS3, 8, 10). At site PS6, close to the eastern border, the grey microgranite is intruded by a 1–2 m thick, subvertical and dark lamprophyre dyke with a N40° strike (PS5).

Restoring the paleohorizontal in granites is generally difficult. Here, the laccolite has first intruded the micaschists upward, along the westerly fault and then, northeastward, parallel to the schistosity. Columnar joints can be seen all along the cross-section. The hanging wall of the laccolite, which is perpendicular to the columns, has a nearly constant dip of 10–15° toward the southwest, whereas the strike varies from N140° (PS10) to the west to N100° to the east (PS2–6) (Fig. 1). At site PS9, the magmatic foliation deduced from the dark ellipsoidic inclusions and from the anisotropy of magnetic susceptibility (AMS) has the same strike as the hanging wall but a steeper inclination of 40°. A similar dip occurs at sites PS3 and 8. At the westernmost site, PS10, the very steep magnetic foliation (N140°, 75°S) confirms that the laccolite is rooted along the western border. On the contrary, at site PS6, which is located close to the eastern border, the mesoscopic foliation seems to be parallel to the cooling surface (N100°, 10–15°S), whereas AMS measurements show a nearly flat magnetic foliation. Neither the upper cooling surface nor the magmatic foliation or columnar joints is a decisive criteria for a paleohorizontal, but they all suggest a south–southwesterly bedding with an unclear dip, likely resulting from a syn- to post-magmatism normal faulting of the NW–SE-striking fault that marks the western border of the laccolite (Fig. 1).

3. The magnetic components

Six to twelve samples per site were cored or taken as hand samples and each core was cut into 2 or 3 specimens. The direction and intensity of natural remanent magnetization (NRM) were measured on a modified Digico spinner magnetometer with a maximum noise level at 0.03 mA/m. Low-field susceptibility and AMS measurements were also done on Digico devices. Isothermal remanent magnetization (IRM) acquisition and thermal demagnetization of three-component IRM (Lowrie, 1990) were performed in order to identify the magnetic phases. Stepwise thermal demagnetization was used to separate the components of the NRM of each specimen. The directions of the components were computed using line fitting to segments in Zijderveld diagrams with at least four points per line (Kent et al., 1983; Kirschvink, 1980). Fold test was applied to the mean directions (McElhinny, 1964; McFadden, 1990; Watson and Enkin, 1993).

3.1. The Doller Valley volcanites

3.1.1. Magnetic carriers

With certain exceptions, IRM procedures and thermal demagnetization of NRM and of the three-components IRM reveal that the carriers of the remanence have low coercivities and maximum unblocking temperatures in the range 500–580°C that are characteristic of magnetite and titanomagnetites with a low content of titanium (Table 1), (Fig. 2: DV1.81, DV9.81, DV8.53, DV13.62; Fig. 3: DV9.22, DV10.42, DV14.111) (Edel et al., 1998; in preparation). At a few sites from the upper layer (DV2, DV5, DV6) were observed lower unblocking temperatures in the

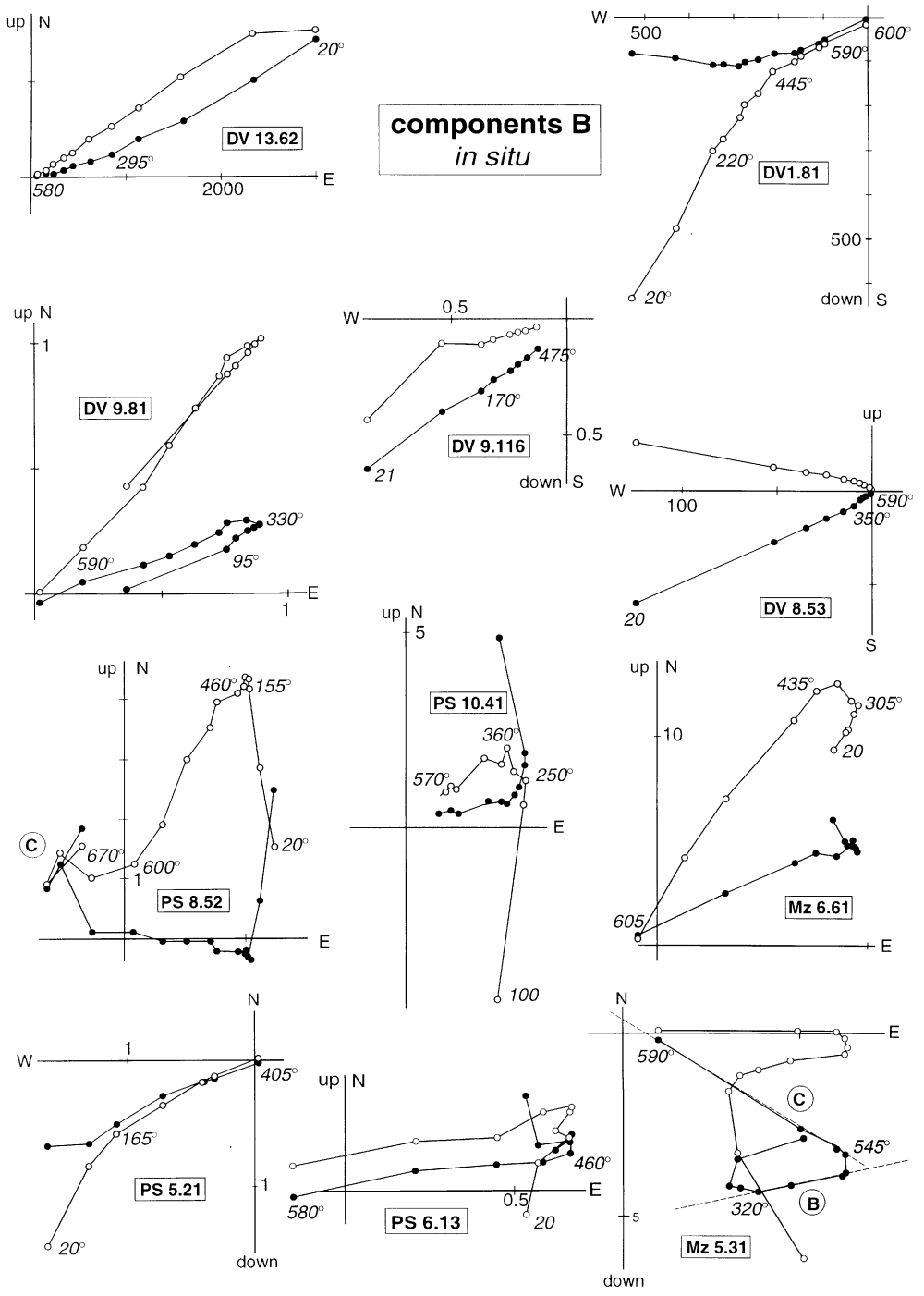


Fig. 2. Typical thermal demagnetization curves, orthogonal projection of *B* components in situ. Full dots are in the horizontal plane, open circles are in the vertical plane. Units are in mA/m.

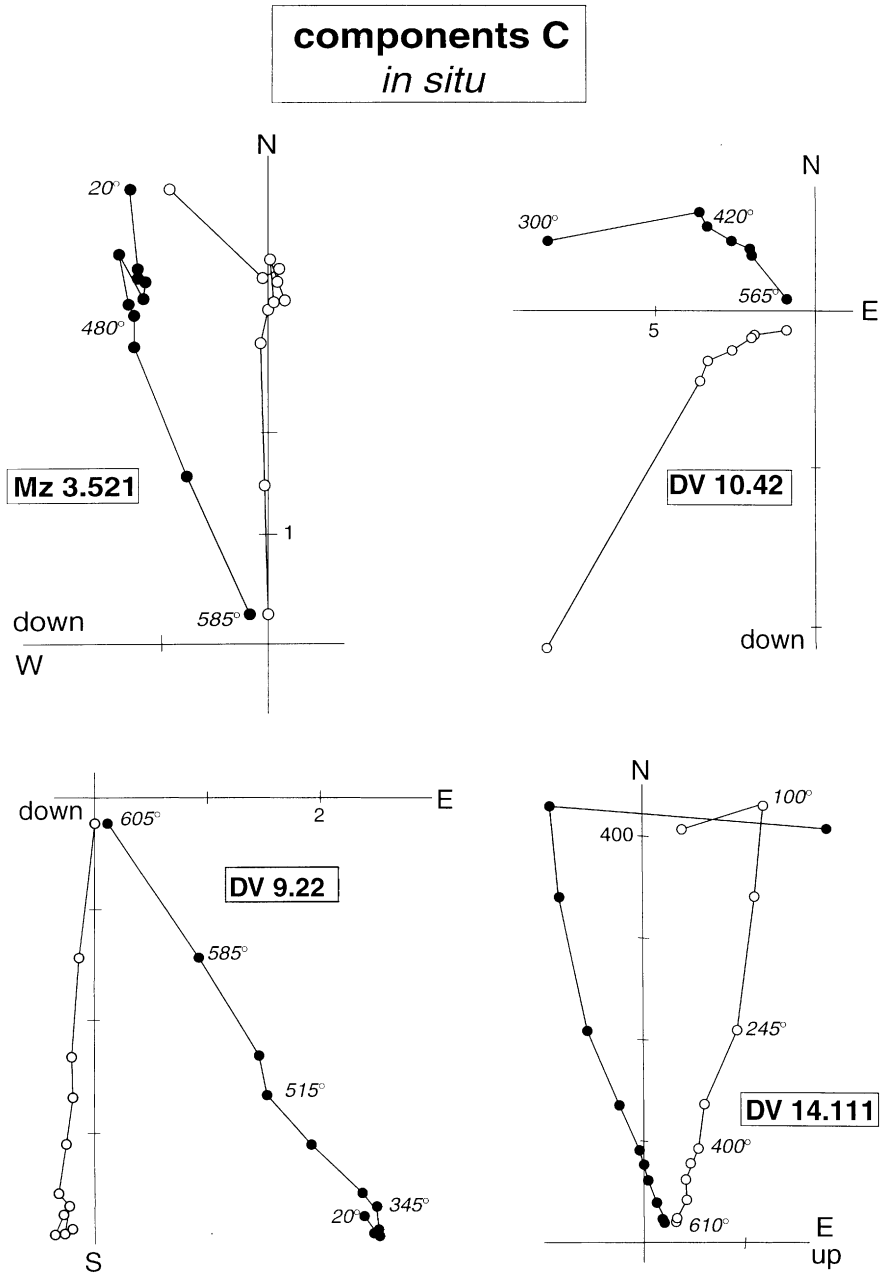


Fig. 3. Typical thermal demagnetization curves, orthogonal projection of *C* components in situ. Full dots are in the horizontal plane, open circles are in the vertical plane. Units are in mA/m.

range 300–400°C. In these rocks we suspect that the magnetization is also carried by magnetite but, due to the low intensities of the NRM (<0.5 mA/m), at 300–400°C, the remaining magnetization cannot be distinguished from the noise. In a separate paper (Edel et al., 1998; in preparation), petrological analysis and rock magnetic procedures show that in the basaltic layer, several generations

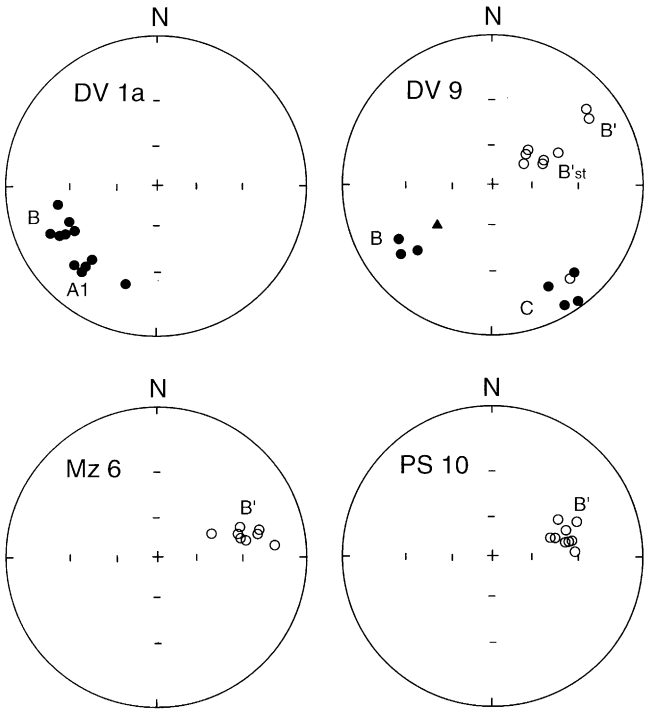


Fig. 4. Stereographic projections of the directions of remanent magnetizations isolated at four sites consisting of: a basalt (DV1) and a pyroclastic unit (DV9) from the southern Vosges, a tuff (Mz6) and a microgranite (PS10) from the Massif Central. The groups of directions are labelled *A*1, *B* (reversed polarity), B' (normal polarity), B'_{st} (syn-tectonic) and *C*.

of magnetite do coexist. At two sites (DV2 and DV6) where part of the rocks show a reddish pigmentation, the IRM curves and unblocking temperatures reveal hematite as carrier of a part of the remanent components.

3.1.2. Remanent components

As expected from the previous studies (Edel, 1987b; Edel and Lewandowski, 1993), the Visean volcanites show a multivectorial behaviour of the remanent magnetization (Fig. 4). At several sites,

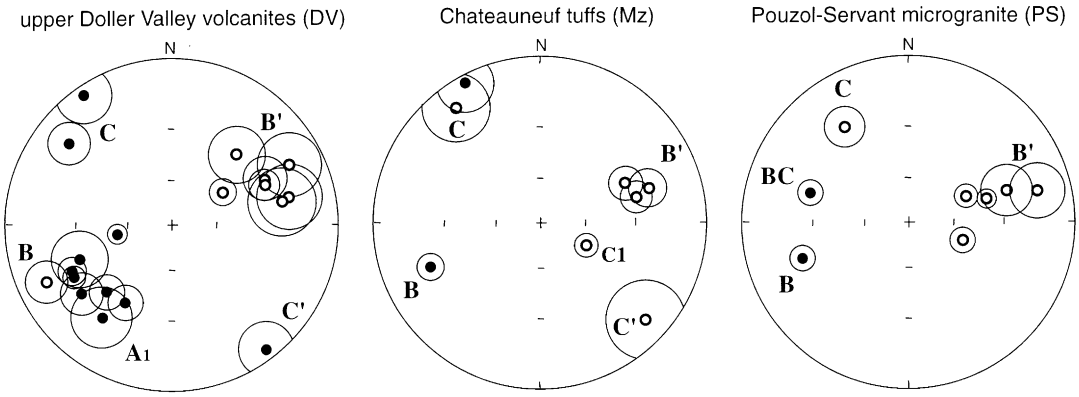


Fig. 5. Stereographic projections of the site mean in situ directions and α_{95} confidence circles. Full/open circles are, respectively, in the lower/upper hemisphere.

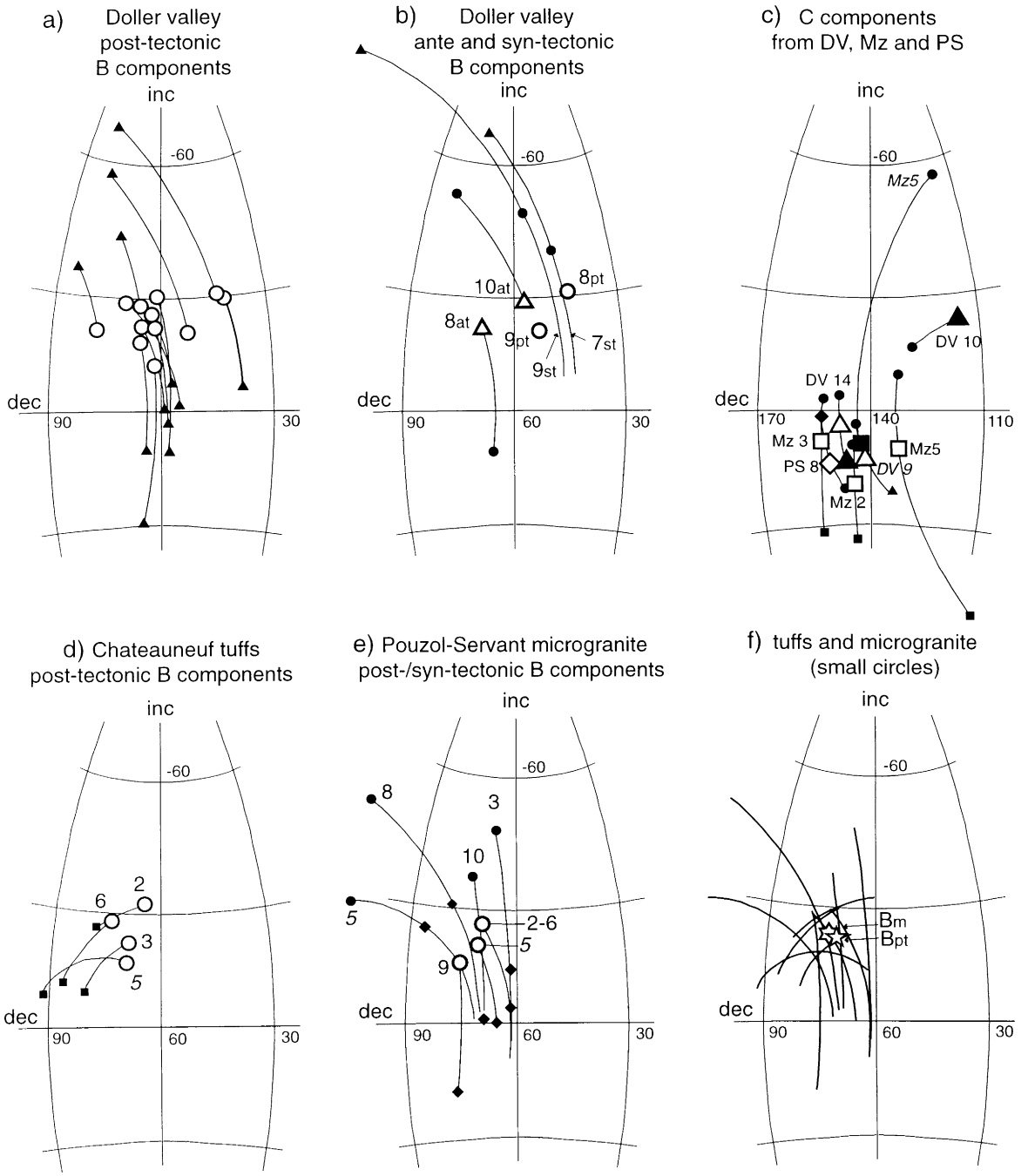


Fig. 6. Equatorial stereographic projection of the mean directions before (circles) and after (triangle, square, diamond) tilt correction and corresponding small circles. All directions with reversed polarity are plotted in the eastern hemisphere together with the direction with normal polarity. The big symbols indicate the directions taken into account for calculations of means. For the microgranite, the two corrected directions are obtained using the dip of the upper cooling surface and the dip of the magmatic foliation. pt: post-tectonic overprint, at: ante-tectonic magnetization, st: syn-tectonic overprint. Stars: calculated mean directions. Bm: mean direction at the minimal distance from the small circles.

post-tectonic overprints, carried by secondary magnetite (DV1b, DV7, DV10, DV12, DV13) or hematite (DV2 and DV6), and interpreted as being Permian to Jurassic in age, were isolated. The directions are not listed here and are the subject of the separate paper (Edel et al., 1998; in preparation). The remaining site mean in situ directions show two main clusters, nearly antiparallel, that are identified as B and B' directions (Fig. 2: DV1.81, DV9.116, DV13.62; Figs. 4 and 5). A few NW and SE mean directions with shallow inclinations correspond to the C directions from previous studies (Fig. 3: DV9.22, DV10.42, DV14.111; Figs. 4 and 5). In order to illustrate the effect of the tilt correction, the normal and reversed directions, before and after tectonic correction, are plotted in the same easterly hemisphere (Fig. 6). At sites DV1a (Fig. 4) and DV13 directions closer to the meridian were found, statistically different from the B and consistent with the $A1$ mean direction from the previous study. After tilt correction they show a significant decrease of the k factor, ($k_c/k_0 = 3/67$). Consequently they are considered as post-tectonic overprints.

3.1.2.1. The post-tectonic B components. For 11 sites, the well-clustered mean B directions do clearly diverge after tilt correction ($k_c/k_0 = 6/48$, Table 1), showing a negative fold test (McElhinny, 1964; McFadden, 1990) (Fig. 5a). The mean directions 239°/28° ($\alpha_{95} = 10°$) and 65°/−24° ($\alpha_{95} = 11°$) computed with the in situ directions with, respectively, reversed and normal polarity can be considered as antiparallel ($\gamma = 6° < \alpha_{95}$). The positive reversal test (McFadden, 1990) supports the reliability of the B magnetizations that are interpreted as overprints acquired after the main Late Visean tilting of the units. Thermal demagnetization and IRM procedures show that magnetite is the main carrier of these secondary magnetizations (Fig. 2). The mean post-tectonic B_{pt} direction for all 11 sites is 62°/−24° ($\alpha_{95} = 7°$) and the mean pole 8°N/126°E (Table 1).

3.1.2.2. Possible pre-and syn-tectonic B components. A few directions do not show the general behaviour seen above. For DV8 (in situ: 246°/−11°) (DV8.53, Fig. 2) and DV10 (in situ 261°/53°) the directions converge toward the previous post-tectonic direction 242°/24° (62°/−24°) as it testifies the mean for both directions 243°/26° obtained after tectonic correction ($k_c/k_0 = 79/3$) (Fig. 6b). This positive fold test implies that the magnetizations were acquired when the units were lying horizontally. The similarity of this direction to others of the group B suggest the same origin, thus it may be regarded as representative component for this group.

Two other exceptions were found at sites DV7 and DV9. Before correction, the declinations are comparable to those of the B components but the inclinations are steeper (B_{st}, Table 1; DV9.91, Figs. 2 and 6b). After correction, the directions diverge and become steeper. From the geology we know that in the internal Variscides, extension has succeeded to the Late Visean tightening. It is likely that at some places, the series have been steeply tilted during the Late Visean transpression and partly back-tilted during the later distension. This solution has already been considered previously for other units of the southern Vosges, in order to explain directions with similar high inclinations, labelled D components (Edel and Lewandowski, 1993). Therefore we suggest that the B_{st} (B syn-tectonic) components were acquired after the Late Visean compression, during the Namurian extension.

3.1.2.3. The C components. The C components from DV9, DV10 and DV14 are generally well defined and display characteristics of Ti-poor magnetites with maximum unblocking temperatures (T_{ub}) close to 580°C (Fig. 3: DV9.22; DV10.42; DV14.111). In the basaltic specimens from DV14 a drop by about 80% of the NRM in the range 100–350°C suggests that the original Ti-rich titanomagnetites are preserved. They were also identified in thin sections. At the different sites of the basaltic layer, the M_S/M_{RS} factor increases and the grain size decreases with the degree of alteration (Edel et al., 1998; in preparation). As at DV14, M_{RS}/M_S is the lowest, the C components represent an old and probably primary magnetization. The inconclusive fold test for the three sites (McElhinny, 1964; McFadden, 1990), indicates that some of the magnetizations are syn- or post-tectonic overprints. It will be shown below that at DV9 and DV10 the magnetizations were acquired after emplacement of the flows.

Table 2
Paleomagnetic results from the Châteauneuf tuffs (Massif Central, 46.05°N/2.90°E). and the Pouzol-Servant microgranite (Massif Central, 46.15°N/2.95°E). See Table 1

Site	Rock type	Bt (°C)	*n*/*N*	*D* (°)	*I* (°)	D_c	I_c	*k*	α_{95} (°)	Comp	VGP (°N)	VGP (°E)
Mz5	Rhyolitic tuff	450–520	8	**249**	**19**	277	13	73	6	B_{pt}	7	115
		520–580	6	133	−9	106	49	17	16	C_{st}(N + R)	−32	62
		300–350, 585	4	116	−57	**143**	**10**	138	8	C_{at}	−29	46
Mz2	Rhyolitic tuff	570–590	13	**65**	**−32**	79	−26	20	9	B_{pt}	4	123
			9	324	−9	323	−34	16	13	C_{st}	−30	46
Mz3	Rhyolitic tuff	570–590	6	**72**	**−22**	81	−9	52	9	B_{pt}	4	114
		580	3	332	3	333	−32	154	10	C_{st}	−39	40
Mz6	Rhyolitic tuff	580–590	8	**75**	**−28**	87	−12	50	8	B_{pt}	−1	114
PS2–6	Microgranite grey	520–580	5	**70**	**−26**	66	−20	37	13	B_{pt} [a,b]	4	112
Ps9	Microgranite grey	520–580	3	**77**	**−13**	76	−2	102	12	B_{pt}		
						80	14	102	12	[b]		
PS3	Microgranite pink	530–580	6	63	−52	62	−42	55	7	[a]		
						60	−12	55	7	[b]		
PS8	Microgranite pink	580	9	110	−49	95	−44	54	7	[a]		
						78	−30	54	7	[b]		
		> 620	11	326	−21	330	−14	26	9	*C*	−38	38
						333	−1	26	9			
PS10	Microgranite pink	580	9	74	−38	72	−29	92	5	[a]		
						70	0	92	5	[b]		
PS5	Lamprophyre	350–400	9	287	28	**281**	**29**	63	6	[a]		
						266	**25**	62	6	B_{at} [b]	−7	105
	Dyke	400–520	7	**252**	**21**	249	15	89	6	B_{pt} [a]	4	113
						246	0	89	6	[b]		
Mean *B*	All		11	77.5	−30			22	10			
						77	−9	20	10			
Mean *B*	Tuffs, grey microgranite and dyke (high temp. comp.)		7	**72**	**−23**			126	5		3	114
						78.5	−4	20	14			
Mean *B*	(Minim. distance to small circles)		11	**74**	**−24**						2	113
Mean *C*	In situ		4	144	4			28	18			
	Corrected		4			142	30	9	30			
	35% unfolding		4			145	13	76	11			
Mean *C*	(35% unfolding + C_{at} of Mz5)		5			**144**	**12**	93	8		−29	45

[a] Direction corrected for the bedding of the hanging wall.
[b] Corrected for the bedding of the magmatic foliation at PS3, 8, 9.

3.2. The tuffs and microgranite

3.2.1. The magnetic carriers

The tuffs of the Châteauneuf area and the microgranite show intensities of the NRM, respectively, in the range 10–150 and 0.1–25 mA/m. The grey and the pink microgranite display contrasting intensities of remanence. In most grey rocks, the NRM does not exceed 0.25 mA/m and at 160°C, the intensity is generally below 0.05 mA/m. As a consequence, for sites PS6, 7 and 9, only few specimens have yielded a reliable characteristic magnetization. Thermal demagnetization of NRM and IRM indicate that, for both rock types, magnetite is the carrier of most characteristic magnetizations (Table 2; Fig. 2: Mz5.31, Mz6.61, PS6.13, PS8.52, PS10.41). Exceptions are a small residual magnetization carried by hematite at PS8 (Fig. 2; PS8.52) and low to intermediate unblocking temperature magnetizations in the tuffs of Mz5 and the lamprophyre dyke PS5 (Fig. 2: PS5.21).

3.2.2. The B components

On the stereoplot, the tuffs exhibit mean B' directions with normal polarity (Mz2, Mz3, Mz6) (Fig. 2: Mz6.61, Fig. 4) and both, normal and reversed polarity B direction at Mz5 (Fig. 2: Mz5.21, Fig. 5; Table 2). The results from the microgranite look similar and reveal predominant B' directions with normal polarity (PS2, PS3, PS6, PS8, PS9) (Fig. 2: PS6.13, PS8.52 and PS10.41). The dyke, which is intrusive into the microgranite, shows a reversed B direction and an additional direction BC intermediate to B and C (Fig. 5).

Monoclinal bedding of both units and an unclear dip of the laccolite preclude a fold test within a same unit. But, the strikes of the tuffs and of the microgranite being nearly perpendicular, a between-site fold test can be done. When using all B components, the test is inconclusive (McElhinny, 1964; McFadden, 1990) ($k_c/k_0 = 20/22$; Table 2). Partial unfolding (Watson and Enkin, 1993) slightly improves the cluster ($k_c = 28$ for 38% unfolding) suggesting syn-folding magnetizing of at least a part of the sites. In fact, the directions of the tuffs, the grey granite and the dyke (intermediate temperature component) show a nice cluster before correction and, according to McElhinny (1964), clearly fulfil negative fold test ($k_c/k_0 = 20/121$) in favour of a post-tilting overprinting (Table 2; Fig. 6d and e). Concerning the reddish microgranite and the low unblocking temperature component of the dyke, the directions converge toward the post-tectonic direction during unfolding, as shown by the small circles (Fig. 6e). The dip correction requisited for best grouping cluster is not the same for the four directions: only 15° for PS9 that is located close to the western border of the laccolite, about 40° for PS8 and PS5 that are at a greater distance of the feeding zone. This means that the hydrothermal activity responsible for the syn-tectonic B magnetizations in the reddish granite was not synchronous at the different sites and that it occurred during tilting of the laccolithe along the western normal fault.

Summing up, the B components of the Late Visean magmatites from the northern Massif Central are essentially post-tectonic overprints, like in the southern Vosges. The mean in situ direction of these overprints, labelled B_{pt}, is 72°/−23° ($k = 121$, $\alpha_{95} = 5$, for $N = 7$; VGP: 3°N/114°E). Pre-to syn-tectonic B components are preserved at four sites. Due to the propituous strike of the tuffs and the laccolite, the small circles of the 11 pre-, syn- and post-tilting B directions do converge. The convergence direction was computed by minimizing the distance between the vector and the small circles. The direction 74°/−24° (VGP: 2°N/113°E) is nearly the same as the mean direction calculated with the in situ directions.

3.2.3. The C components

The C directions are well defined but scarce at Mz3 (Fig. 3: Mz3.521; Table 2). At Mz2, the C components show lower maximum unblocking temperatures than the B in part of the specimens and slightly higher maximum unblocking temperatures (580–590°) in others, indicating that the C components are carried by magnetites with various grain sizes whereas the later B components are mostly carried by small grains. After removing the B components from Mz5, at 520–570°, well defined components remain that are eliminated at 580–600°. These directions show three clusters, one C with normal polarity and shallow inclination sample (Mz5.31, Fig. 2), one in the opposite direction and one $C1$ with high inclinations (Fig. 5). The carrier of the C components is magnetite in the tuffs (Mz2, 3, 5) and hematite in the pink microgranite of PS8. In this latter site, a phenomenon that we met

Table 3
Paleomagnetic results published for Early Carboniferous units from the western Europe Variscides. Location of the investigated areas on Fig. 1. k_{is}: Fisher statistic parameter k before tilt correction, k_c: same after correction, k_m: same for partial unfolding. C_{n2} (**) is a subgroup of C_n (Fig. 8)

Area	°N/°E	Rock formation	Rock age	N	Tect. setting	D/I	k_{is}	k_c	k_m	α_{95}	VGP (°N/°E)	Pol	References
N. Massif Central (Mt)	46.3/2.8	Diorite, granite	Videan?	2	C_n in situ	121/29	**54**				−9/59	N	Edel, 1987a,b
N. Vosges (NV)	48.4/7.2	Diorites, grandodiorites	Visean, 345–340 Ma	11	C_n in situ	326/−28	**44**			7	−20/42	N + R	Edel and Wickert, 1991; in preparation
N. Vosges (NV)	48.4/7.2	Sediments	Devonian–Visean	3	C_n in situ	318/−25	**31**	**14**		22	−**18**/50	N + R	Edel and Wickert, 1991; in preparation
S. Vosges (SV)	47.8/7.0	Volcanites	Visean, 345–336 Ma	9	C_n unfolded	138/39	9	**42**		10	−**11/46**	N + R	Edel and Lewandowski, 1993
Mean C_n				4					**88**	**10**	−**14/49**	N + R	
Spessart (Sp)	50.0/9.2	Metam. rocks, diorites	Visean	5	C_o in situ	327/−11	**78**			9	−27/47	N	Edel and Wickert, 1991
N. Vosges (NV)	48.4/7.2	Diorites, granodiorites	Visean, 345–340 Ma	5	C_{n2} in situ	328/−18	**129**			7	−26/43	N	**
S. Vosges (SV)	47.8/7.0	Volanites	Visean, 345–336 Ma	12	C_o 60% unfolded	144/4	**17**	19	**23**	9	−31/50	N + R	Edel and Lewandowski, 1993
S. Vosges, N. Mass. Central		Volcanites, microgranite	Visean, 345–340 Ma	7	C_0 unfold. +35% unfold.	(145/12)			**139**	5	−**28/46**	N + R	This study
Mean C_0				**4**					**568**	**4**	−28/46	N + R	
N. Vosges (NV)	48.4/7.2	Diorites	Visean, 345–340 Ma	3	C_p in situ	326/32	54			17	−48/61	R	Edel and Wickert, 1991
N. Vosges (NV)	48.4/7.2	Sediments	Devonian–Visean	6	C_p 40% unfolded	320/28	36	30	**45**	10	−43/66	N + R	Edel and Wickert, 1991
S. Vosges (SV)	47.8/7.0	Volcanites	Visean, 345–336 Ma	16	C_p in situ	136/−26	**19**	5		8	−40/69	N + R	Edel and Wickert, 1991
S. Vosges (SV)	47.8/6.9	Diorites, monzonite	Visean, 342 Ma	9	C_p in situ	133/−26	**28**			10	−38/72	N + R	Edel and Wickert, 1991
S. and M. Vosges	47.8/7.0	Dykes	330–328 Ma	3	C_p in situ	122/−22.5	**77**			14	−30/80	N + R	Edel and Wickert, 1991 and 1 new data
Mean C_p				**5**					**89**	**8**	−**40/70**	N + R	
NE Massif Central (Ro)	45.9/4.2	Tuffs, ignimbrites	Visean	6	*B* 50% unfolded	75/−14	20	20	**34**	12	5/110	N	Edel et al., 1981
				3	*B* unfolded*	62/−4	1	**27**		24		N + R	Edel et al., 1981
N. Massif Central (Co)	46.3/2.0	Doprotes	Visean	2	*B* in situ	230/21	87				17/130	R	Edel, 1987a,b
N. Massif Central (Mz–Ps)	46.1/2.9	Tuffs, microgranite	Visean, 340 Ma	7	*B* in situ	72/−23	121			5	3/114	N + R	This study
S. Vosges basin (SV)	47.8/7.0	Volcanites	Visean, 345–336 Ma	13	*B* in situ	66/−27	**44**	5		6	5/124	N + R	Edel and Lewandowski, 1993
S. Vosges bains (DV)	47.8/6.9	Volcanites	Visean, 345–340 Ma	13	*B* 11 in situ, 2 unf.	62/−25	**58**			5	8/127	N + R	This study
N. Vosges (NV)	48.4/7.2	Rhyolites, hostrocks	Visean, 331–328 Ma	5	*B* in situ	54/−21	**153**			6	14/132	N	Edel and Wickert, 1991
S. Black Forest (Mü)	47.9/7.8	Rhyolitic ignimbrites	Visean, 33 Ma	3	*B* unfolded*	66/−22	57	**122**		11	7/124	N	Edel, 1987b
Odenwald (Od)	49.7/8.7	Diorites, grandodiorites	Visean, 340 Ma (Hb), 330 Ma (Bt)	6	*B* in situ	64/−27	60			9	5/127	N + R	Edel and Wickert, 1991
Mean *B*				**7**					**80**	**6**	**8/123**	N + R	

* Unfolding only the Tertiary tilt.

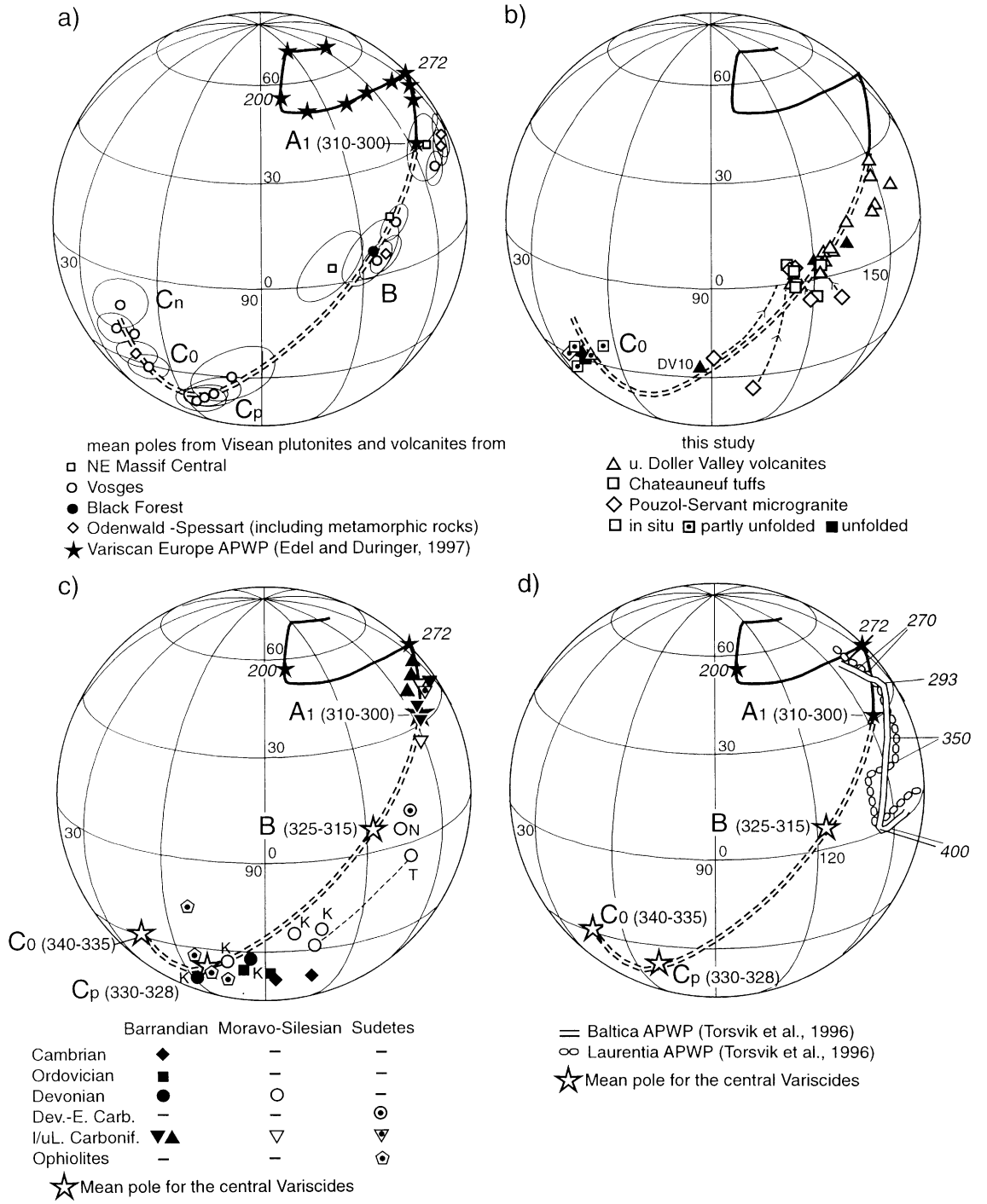


Fig. 7. Orthogonal projection of: (a) the mean poles from previous studies on Visean units of the Central Variscides (Table 3) and the APWP of Variscan Europe (Late Carboniferous–Jurassic) (Edel and Schneider, 1995; Edel and Duringer, 1997); (b) the poles from this study; (c) poles published for the Bohemian Massif (Mauritsch and Pruner, 1990; Nawrocki, 1993 [N]; Krs and Pruner, 1995 [K]; Tait et al., 1996 [T; the dashed line joins the poles of a bimodal distribution]) and the mean poles from the western massifs (Table 3); and (d) the calculated APWPs of Baltica and Laurentia and the mean poles of the central Variscides.

on some other occasions can be observed. Due to mineralogic transformations during heating, also evidenced by an increasing susceptibility above 580–600°C, the small residual magnetization is enhanced, and its direction scatters around a mean *C* direction (PS8.52, Fig. 2). The directions of this residual component are nearly the same for all specimens ($\alpha_{95} = 9°$ for 11 specimens), so that it can be considered as characteristic for the site (PS8, Table 3; Fig. 5).

The inconclusive to negative fold test (McElhinny, 1964; McFadden, 1990) obtained with the four mean directions with shallow inclinations ($k_c/k_0 = 28/9$) excludes a primary acquisition of the magnetization. Partial unfolding by 35% (Watson and Enkin, 1993) significantly improves the cluster around the mean 145°/13° ($k_c = 76$, $\alpha_{95} = 11$), indicating that the magnetization is syn-tectonic. After tilt correction, the *C*1 component with high inclination (Fig. 5), found in the freshest, fine-grained specimens of Mz5, has a direction 143°/10° remarkably close to the previous syn-tectonic direction. This means that this component is the sole pre-tectonic and perhaps primary *C* magnetization of the tuffs.

4. Discussion

4.1. Comparison with results from the western Variscides

4.1.1. The C components (Table 3; Fig. 7)

The previous studies have shown that: (1) *C* directions are the most abundant in the Visean volcanic and intrusive units from the Vosges; and (2) they display a trend from C_n (NW with negative inclinations around −35°, or conversely) to C_p (NW with positive inclinations around 25°, or conversely) without significant differences in declination (Edel and Wickert, 1991; Edel and Lewandowski, 1993). These *C* directions can be divided into four groups depending on their response to tectonic correction.

Late Visean, rhyodacitic and rhyolitic tuffs and ignimbrites from the southern Vosges basin, dated at 342–340 Ma by U–Pb chronometry (Schaltegger et al., 1996) and generally located far from the intrusive monzonitic and dioritic Ballons massif, show a positive fold test ($k_c/k_0 = 24/9$ for 10 sites, Table 3) and a C_n direction 139°/36° (VGP: 13°S/46°E) that was interpreted as pre-tectonic (Edel and Lewandowski, 1993). A close C_n direction, but with a mainly opposite polarity 326°/−29° ($k = 40$, $\alpha_{95} = 8°$, $N = 10$; VGP: 19°S/42°E), characterizes the dioritic and granodioritic rocks of the Champ du Feu massif (northern Vosges), that were emplaced at the same period as the volcanites and plutonites of the southern Vosges. A ^{39}Ar–^{40}Ar plateau on biotite indicates a cooling age of 340 Ma for a granodiorite carrying a typical C_n component (Boutin et al., 1995; VN 53 in Edel and Wickert, 1991).

In the southern Vosges basin, the most important group is characterized by C_p directions and a clear negative fold test (McElhinny, 1964) (Table 3). The mean direction 136°/−26° (VGP: 40°S/69°E) is nearly the same as the mean 133°/−26° (VGP: 38°S/72°E) obtained from various dioritic intrusives from the southern Vosges (Edel and Wickert, 1991; Table 3). The next group consists of directions that show statistically optimal C_p cluster after partial untilting (Watson and Enkin, 1993) (Table 3). The mean direction 320°/8° (VGP: 43°S/66°E) also fits the two previous means. The last group, labelled C_o, shows an inconclusive fold test, but the inclinations are intermediate to C_n and C_p. Here again, partial unfolding by 60% improves the grouping (Table 3). The mean direction is 144°/4° for 12 sites (VGP: 31°S/50°E).

Actually, the major part of the *C* components from the Vosges consist of overprints, acquired partly before, partly during and partly in the latest stages of tilting. This is not astonishing when considering the tectonic evolution of the area (Schneider, 1990, 1994; Schulmann et al., 1997) and the radiometric ages (Schaltegger et al., 1996, 1997). Folding began during the Late Visean volcanism and was associated with granitic intrusions dated at 340 Ma (Petrini and Burg, 1998) and reached a peak subsequent to the emplacement of the latest volcanic-sedimentary units. Between 340 and 333 Ma, rapid uplift brought the deep metamorphic units of the central Vosges and Black Forest close to the surface. From 333 to 328 Ma, syn- to post-tectonic granites and dykes intruded the basement. This 10–12-Ma-long tectono-magmatic activity was the context of acquisition of the *C* components and explains their quite complex behaviour.

The youngest rocks characterized by typical C_p

components are mafic dykes intruded into the Ballons massif and into migmatites of the middle Vosges (Table 3). The mean direction 122°/−22.5° (VGP: 30°S/80°E) looks similar to the direction 133°/−26° found in the 342–340-Ma-old hostrocks of the Ballons massif, but is slightly deviated. The position of the pole, between the other C_p and the B poles, indicates that the magnetizations in the dykes are the youngest of the C_p group. One of the dykes of the Ballons massif has provided a $^{39}Ar–^{40}Ar$ age of 329 ± 2 Ma on biotite (Boutin et al., 1995) and the migmatite that contains the dioritic dyke was recently dated at 328 ± 4 Ma by U–Pb chronometry (Schaltegger et al., 1997). The close C_p directions from the dykes and from the dioritic hostrocks suggest that the latter have been remagnetized during the late magmatic event. Actually, the numerous $^{39}Ar–^{40}Ar$ and K–Ar dates around 330 Ma found in the intrusive and metamorphic units of the southern and central Vosges (Boutin et al., 1995) argue in favour of a resetting as well of the magnetizations as of the ages, due to uplift and/or the late magmatic intrusions. Consequently, we consider that the C_p components were acquired during the late stage of the Visean tightening, i.e. around 330 Ma. The pre- to syn-tectonic C_o magnetizations were acquired during and after the volcanism at 342–340 Ma and before 333 Ma, which is the age of the ignimbrites that cover the uplifted metamorphic core of the Black Forest (Schaltegger, 2000).

Excepted at site PS8, the C components exhibited from DV9, DV10, DV14, Mz2, Mz3 and Mz5 are well-defined vectors with maximum unblocking temperatures in the range 575–600°C and IRM characteristics of magnetite. Remnants of titanomagnetite may be preserved at DV14. The directions from the Massif Central and Vosges look very similar (Fig. 6c). With the exception of the component with the high inclination of Mz5 that is considered as pre-tectonic, the C components from the Massif Central are syn-tectonic overprints. The excellent consistency of the VGP (27°S/46°E) of the corrected direction of DV14 with the mean VGP (29°S/45°E) obtained from the tuffs and microgranite of the Massif Central confirms the hypothesis of a primary C magnetization deduced from magnetic mineralogy (Edel et al., 1998; in preparation). The corrected C pole of DV10 26°S/87°E is located on the APWP of the Vosges (in this paper the concept of APWP that is usually understood as a chain of poles corresponding to primary and dated magnetizations is extended to poles calculated from overprints). The pole is not far from the mean C_p pole 38°S/72°E previously obtained in the diorites and monzonites of the Ballons massif and very close to the C_p pole 30°S/80°E yielded from dykes that have intruded the dioritic hostrocks around 330 Ma (Table 3; Fig. 7a and b). The vicinity of the DV10 volcanites with the monzonite and the consistency of the result with the late C_p directions highly suggests that this direction is a C_p overprint acquired during the late thermo-magmatic event and tilted together with the series during uplift of the granitic massif. With regards to DV9, the in situ direction is close to the corrected direction of DV14, but partial unfolding makes them lying closer. The optimal cluster of the C VGPs is obtained after unfolding by 35% for the directions of Mz2, Mz3, Mz5, PS8 and DV9 and by 100% for the directions of Mz5 (high-inclination component) and DV14. The mean pole (28°S/45°E, $k = 139$, $\alpha_{95} = 5°$) fits the previous C_o VGP (31°S/50°E) of the southern Vosges that also consists of predominantly syn-tectonic overprints.

C_n and C_p directions were also found in dioritic inclusions of the Montmarault granite and in Visean tuffs of the Combrailles (northern Massif Central, Fig. 1) (Edel, 1987a,b). A recent investigation carried out in the Limousin (southwestern Massif Central) in order to evidence the major hydrothermal phases has yielded a large set of C directions in Visean granites as well as in the metamorphic hostrocks (Edel, 1999). In the Saxothuringian zone, a diorite and metamorphic units of the Spessart exhibit directions consistent with the C_o directions from the Vosges (Fig. 1) (Edel and Wickert, 1991).

The change in inclinations of the C directions can derive from an incomplete unfolding and/or from a latitudinal drift of the investigated massifs. In the latter case, the C_n–C_p segment of poles implies an important motion of about 3000 km and a velocity of more than 30 cm/year that is two times greater than the fastest velocities observed in recent times. This signifies either that the rheology of the mantle was not the same as that of the present, or that the C_n–C_p segment is, at least partly, due to ill-unfolded directions. The negative fold test of the C_p directions from the Vosges and, as it will be shown in future papers,

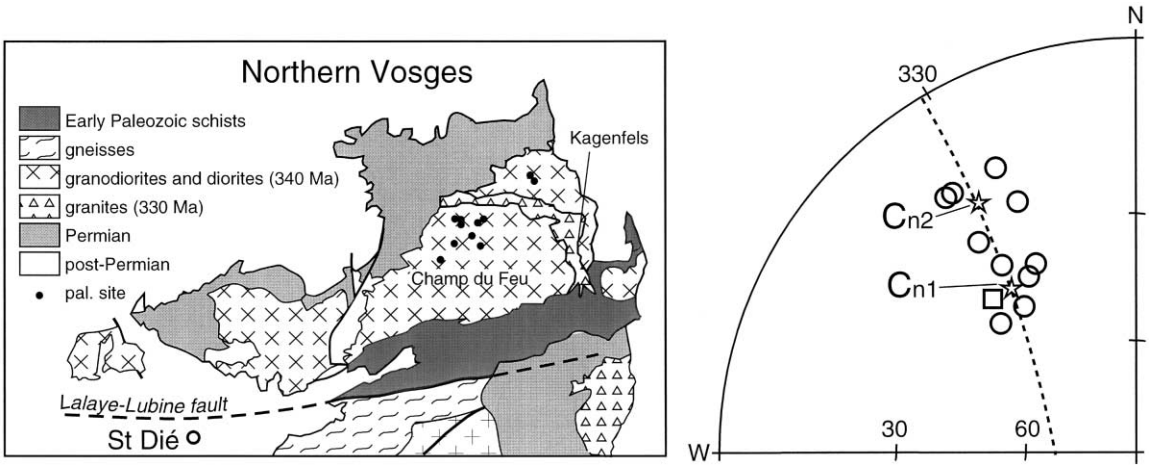


Fig. 8. Geological sketch map of the northern Vosges, location of the sites characterized by C_n directions and stereographic projection of the mean site directions C_n. C_{n1} and C_{n2} are the mean directions corresponding to the poles of the bimodal distribution. These directions are located on a small circle (dashed line), whose axis is horizontal and parallel (N80°) to the intrusions and to the major structural feature of the area, the Lalaye-Lubine fault.

the consistency of the C_p poles through the whole Variscides, allow us to consider the C_p inclinations as representative of the paleolatitude around 330 Ma. On the contrary, the possibility that the C_n directions are not primary but tilted directions of overprints exists. The elongated distribution of the mean C_n directions obtained in the northern part of the dioritic and granodioritic Champ du Feu massif (northern Vosges) suggests a syn-tectonic acquisition of the magnetizations (Fig. 8). The two poles C_{n1} (323°/−37.5°; VGP: 13°S/43°E) and C_{n2} (328°/−18.5°; VGP: 26°S/43°E) of the distribution are located on a small circle whose axis is horizontal and strikes parallel to the dioritic intrusions and to the major Lalaye-Lubine fault that separates the Saxothuringian northern Vosges from the Moldanubian central Vosges. This implies that cooling of the intrusives and acquisition of the C_n components occurred during tilting of the massif. As a consequence, the Lalaye-Lubine fault was not only a dextral strike-slip fault around 340 Ma (Edel and Weber, 1995), but also a normal fault. The consistency of the VGP (26°S/43°E) of the C_{n2} direction with the C_o VGPs of this study (28°S/45°E), of the previous study in the southern Vosges (31°S/50°E; Edel and Lewandowski, 1993) and with the result from the Spessart (27°S/47°E; Edel and Wickert, 1991) indicates that the C_{n2} components are younger than those of the C_{n1} and argues in favour of their reliability in terms of paleogeography. The deviated C_{n1} directions result from a southward tilting by about 20° of the Champ du Feu massif in relation with updoming of the metamorphic and granitic central Vosges, south of the Lalaye-Lubine fault. As a consequence the C_n directions from the southern Vosges can no more be considered as primary, despite of the positive fold test. The magnetizations were acquired in units with similar monoclinal bedding and dispersed by later tectonics. To clear this problem of the C_n directions, further investigations in the Massif Central and the Bohemian Massif are in course. At the present stage of knowledge we consider the C_o–C_{n2} directions as the most representative of the paleogeography of the central Variscides around 340 Ma.

4.1.2. The B components (Table 3)

The mean post-tectonic B direction 62°/−24° (VGP: 8°N/126°E), exhibited from the Middle Visean volcanites of the southern Vosges (this study), agrees well with the direction 66°/−27° (VGP: 5°N/124°E) previously obtained from Late Visean tuffs, ignimbrites and sediments from other parts of the basin (Table 3) (Edel and Lewandowski, 1993). The negative fold test ($k_c/k_0 = 6/46$ for $N = 11$ sites) looks also very similar to the previously published ($k_c/k_0 = 6/44$ for $N = 13$). The secondary acquisition

of the magnetization, deduced from this negative fold test and from a negative conglomerate test (Edel and Lewandowski, 1993), is confirmed here. Nevertheless, the present study suggests that a few *B* components were acquired during late extension tectonics.

On the other side of the Rhinegraben, in the central and southern Schwarzwald, a direction similar to *B* was obtained in the Münstertal ignimbrites. These pyroclastic rocks were emplaced on the uplifted metamorphic basement around 333 Ma (Schaltegger, 2000) and are considered as equivalent to the Latest Visean ignimbrites from the southern Vosges (Edel, 1987b). Here the unit has a strike parallel to the neighbouring Rhinegraben. After correcting the bedding of probable Tertiary age, the mean direction 66°/−21° (VGP: 7°N/123°E) for the three sites is nearly the same as for the *B* components in the southern Vosges.

In the northern Vosges, the 330-Ma-old Kagenfels granite (Hess et al., 1995), intruded the 10 My older granodioritic Champ du Feu massif that is characterized by a predominance of C_n directions and some C_p (see below and Table 3). The *B* components with a mean direction 54°/−21° (VGP: 14°N/132°E) carried by hematite were exhibited in the rhyolitic extremity of the granite as well as in dioritic hostrocks (Edel et al., 1986). These magnetizations are clearly younger than the C_n. They were supposed to be primary, but a recent investigation on an additional site in the central part of the rhyolite shows a predominance of C_p directions. This finding is in favour of a secondary acquisition of the *B* components. More to the north, calc-alkaline intrusives of the Odenwald (Fig. 1) comparable to those of the northern Vosges and also emplaced around 340 Ma (Kreuzer and Harre, 1975; Krohe, 1992), have exhibited a mean *B* direction 64°/−27° (VGP: 5°N/127°E; Edel and Wickert, 1991). For these rocks, the K–Ar ages on biotite indicate that the rocks cooled below 300°C around 330 Ma (Kreuzer and Harre, 1975).

The anthraciferous tuffs of the Roanne basin (eastern Massif Central) result from the same volcanic episode as the Chateauneuf tuffs (Edel et al., 1981; Schneider et al., 1989). The fold test is inconclusive. Part of the investigated sites show a bedding with a strike parallel to the Tertiary, NNW–SSE-striking Limagne graben, while the other displays a Variscan, E–W to SW–NE strike. Partial unfolding of the results from the latter sites (Watson and Enkin, 1993) provides the best *B* cluster around the mean 75°/−15° (VGP: 5°N/110°E) that is close to the result from this study (Table 3).

So, the *B* components can be, in general, considered as overprints acquired after the main Variscan transpressive phase that occurred in the time range 340–330 Ma. At a few sites of the Doller Valley ($DV7_{st}$ and $DV9_{st}$; Fig. 6b) and of the Pouzol-Servant microgranite (the red granite of PS3, PS8 and PS10) we found some arguments suggesting syn-tectonic *B* overprinting. Back-tilting in the first case and normal faulting of the Pouzol-Servant microgranite in the second case indicate that the *B* overprinting was contemporaneous with a post-Late Visean extensive phase. A very similar result in favour of an acquisition of a part of the *B* components during the Namurian extension was recently obtained in the southern Limousin (SW Massif Central) (Edel, 1999).

Due to the lack of primary *B* magnetizations, we can only propose limits for the age of acquisition of the overprints. They are younger than the youngest rocks carrying C_p magnetizations, i.e. the 330–328-Ma-old dykes and Kagenfels rhyolite, and older than the oldest rocks characterized by *A*1 magnetizations. The rocks that are carrying the latter are granites that intruded the Central Variscides at 310–300–Ma (Edel and Lewandowski, 1993). Consequently, the *B* magnetizations have been acquired in the time range 328–310 Ma.

4.2. Comparison with results from the eastern Variscides

In the eastern part of the Variscan belt, reliable Early–Middle Carboniferous data are lacking. NRM from Early–Middle Carboniferous plutonites of the southern part of the massif (Waldviertel–Mühlviertel) and Early Carboniferous sediments of the Franconian Forest consist only of Late Carboniferous and Permian overprints (Reisinger et al., 1994; Bachtadse et al., 1983). Nevertheless, it seems that the Pre-Late Carboniferous paleomagnetic signature has been recorded in older units.

A review of the data available for Cambrian to Devonian units shows that a part of the published VGPs are more or less consistent with the Carboniferous

poles of the Vosges and Massif Central (Fig. 7c). It is the case for:

(1) Various outcrops of ophiolitic rocks and of Devonian sediments of the Polish Sudetes, characterized by poles that are consistent with the *C* and *B* poles of the western massifs as well as by poles corresponding to later overprints (Jelenska et al., 1995; Edel et al., 1997). A part of the Devonian sites which show typical *B* poles are metamorphosed at the contact with Early–Middle Carboniferous granites and consequently, are magnetically overprinted.

(2) Late Cambrian volcanites, Early Ordovician sediments and tuffs, Devonian limestones of the Barrandian sequence, that all display poles in the close vicinity of the mean C_p pole (Krs and Pruner, 1995; Mauritsch and Pruner, 1990; Krs, oral communication).

(3) Devonian sediments of the Moravo-Silesian platform that yielded poles also close to the mean C_p and *B* poles of the western massifs (Krs and Pruner, 1995; Nawrocki, 1993; Tait et al., 1996). The recent result of Tait et al. (1996) obtained from Devonian sediments, shows a bipolar distribution of magnetizations supposed to be primary (T: Fig. 7c), with one half that has a mean pole close to the pole from the Upper Silesian basin (Nawrocki, 1993) and to the *B* poles of the Sudetes (Edel et al., 1997), and the second half that fits the results of Krs and Pruner (1995) (K: Fig. 7c) and draws nearer to the *C* poles of the Vosges and Massif Central.

This consistency of a part of the "Cambrian-Ordovician" poles and of the "Devonian" poles with the Carboniferous C_p and *B* poles of the western Variscides leads to two possible interpretations: (1) the magnetizations have the ages of the investigated rocks and the Bohemian Massif underwent in the Late Devonian the same *C*–*B* and *B*–*A*1 rotations as did the western massifs in the Middle–Late Carboniferous; and (2) the magnetizations are not primary but Carboniferous overprints, and the geotectonic history of the western and eastern parts of the belt is the same. According to the main Carboniferous tectono-magmatic features that occurred in the Bohemian Massif, the second hypothesis seems to be the most likely. The Tepla-Barrandian Proterozoic terrane with its Paleozoic sequence is regarded as a pop-up structure (Edel and Weber, 1995) that, in the context of the Early–Middle Carboniferous (340–330 Ma) transpression, subsided into the "hot and soft" root of the Variscides (Schulmann et al., 1997), while intrusion of granites along the boundaries, rapid uplift, evolving into northward thrusting of the metamorphic units in the north of the structure and southward thrusting in the south, took place. During the same period, i.e. 340–327 Ma, the Moravo-Silesian platform has obliquely underthrusted the Moldanubian zone (Dallmeyer et al., 1992; Stipska and Schulmann, 1995). Later, around 310 Ma, the Moravo-Silesian basement was deformed, mainly along the transcurrent boundary with the Moldanubian (Matte et al., 1990). A similar evolution and a similar timing is observed for the Vosges and the Massif Central (Schulmann et al., 1997). As the same causes produce the same effects, magnetic overprints in relation with uplift and magmatism can be expected. The results published for the Devonian and Ordovician listed above may be interpreted in this way when considering the vicinity of the Barrandian with the Central Bohemia pluton as well as the neighbourhood of the Devonian of the Moravo-Silesian basement with the south Bohemian thrust.

Unless new results on Early Carboniferous intrusives of the Bohemian Massif indicate, we will consider that the paleomagnetic and consequently, the geotectonic evolution of the western and eastern part of the Variscides in the Carboniferous were the same. At the easternmost extremity of the Armorican units, in the Holy Cross Mts (HCM) and upper Silesian massif (USM), a systematic deviation of the VGPs obtained for the Devonian sediments, from the APWP of Baltica has lead Lewandowski (1993, 1994) to contest the Devonian age of the magnetizations (Nawrocki, 1993) and to include the HCM and USM in a similar model of geotectonic evolution.

4.3. The collision of the Variscides with Laurussia

4.3.1. The APWPs of Laurussia

The poles of the central Variscides, Baltica and Laurussia (Torsvik et al., 1996) first converge toward the *B* poles area (Fig. 7d). Then they continue close together, up to the upper Carboniferous–Permian, where they join definitively, suggesting that the history of the Variscides and northern Europe was common since acquisition of the *B* components in the Variscides, probably at 325–320 Ma. But, when

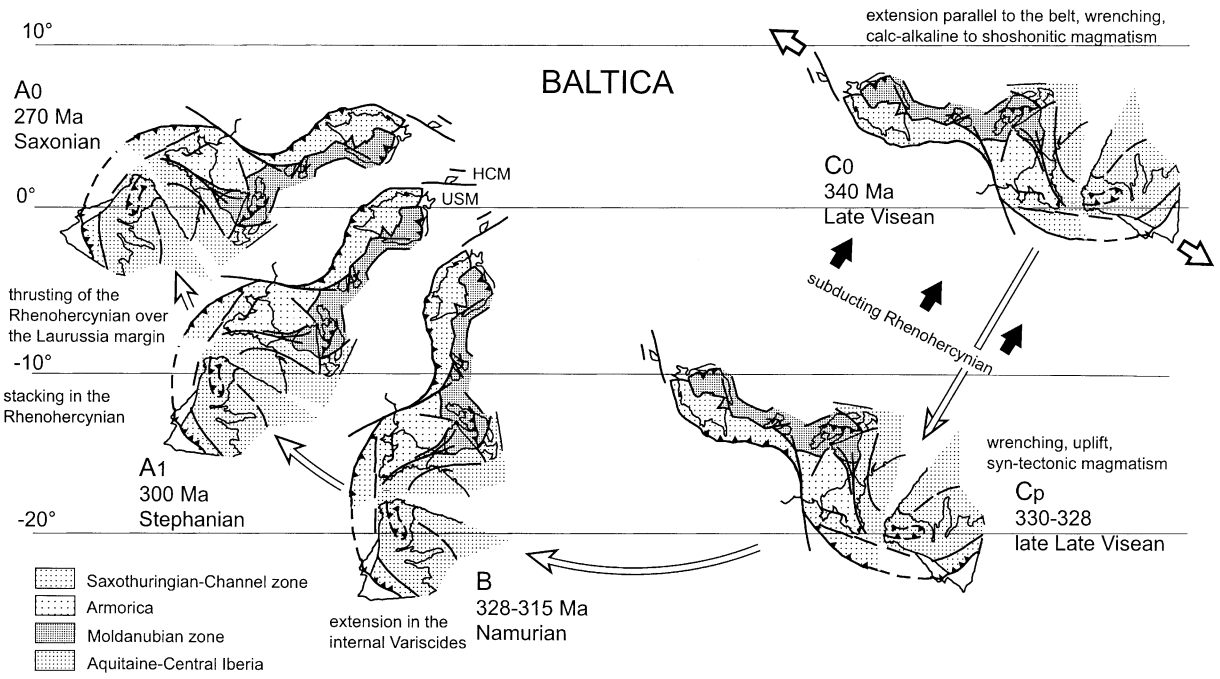


Fig. 9. Paleogeographic evolution of the Variscides from the Early Carboniferous to the Mid-Permian and main tectonic features. HCM: Holy Cross Mountains; USM Upper Silesian Massif. Deformations within the belt are not represented.

we consider the ages of the calculated poles, then the APWPs become common only in the Latest Carboniferous–Permian, around 290 Ma.

The Silurian–Early Devonian poles of Baltica, Britain and Laurentia (Torsvik et al., 1996; Van der Voo, 1990) are consistent, implying that Laurussia was already formed. These poles are not very far from the Middle Carboniferous *B* poles of the Variscides (Fig. 7c). To reconstruct the paleogeography of the Variscides relative to Laurussia during the time-range concerned by this study, reference data around 340, 330 and 325–320 Ma are needed. Such data are lacking for Baltica and scarce for Laurentia. The reconstructions and their geological consequences require a thorough discussion for which there is no place here. Therefore we show the paleogeographic evolution of the Variscides alone (Fig. 9) and produce the complete reconstructions in a forthcoming paper. In the reconstructions the position of Baltica derived from the APWPs of Baltica and Laurentia (Torsvik et al., 1996) is only suggested.

4.3.2. The collision

In the Emsian–Middle Devonian, an oceanic basin generally considered as minor, the Lizard–Giessen–Harz basin, separated the Rhenohercynian Avalonia from the Saxothuringian microplate. Closure of this basin by subduction under the Saxothuringian continental crust started in the Late Devonian. Arc magmatism dated at 340 Ma (Kreuzer and Harre, 1975; Edel et al., 1986; Boutin et al., 1995) developed in the northern Vosges, the Odenwald and the Spessart (Krohe, 1992; Oncken, 1998). The complete closure of the basin and the collision of the Saxothuringian fore-arc and arc with the Rhenohercynian passive margin occurred at the beginning of the Namurian, according to K–Ar ages at 327–325 Ma on phengite of the high-pressure metamorphism recorded along the plates boundary (Ahrendt et al., 1983).

Within the Rhenohercynian foreland, stacking lasted from 320 to 300 Ma (Ahrendt et al., 1983; Weber and Behr, 1983) with cooling ages in the Ardennes at 310 Ma (Spaeth et al., 1985) and 316 Ma (Piqué et al., 1984). Shortening and thrusting

over the London–Brabant basement (Matte and Hirn, 1988; Oncken et al., 1999) lasted up to the latest Carboniferous. At the Variscan front, the northern border of the Dinant syncline attained anchizonal metamorphic conditions around 297 Ma (Piqué et al., 1984).

The closure of the Lizard–Giessen–Harz ocean and the collision of the Saxothuringian and Rhenohercynian zones marks the end of the C_p–B rotation. As the youngest rocks carrying the C_p components are dated at 330–328 Ma, the rotation must have been rapid. The B overprinting affected the internal zones of the Variscan belt after the collision. The significant distance of the B poles of the Mid-European Variscides with the calculated Middle Carboniferous poles of Laurentia and Baltica (Fig. 7d) involves an additional clockwise rotation of the Variscides relative to Laurussia by about 45° in the Middle–Late Carboniferous. The cooling ages at 310–316 Ma recorded in the Ardennes (Piqué et al., 1984; Spaeth et al., 1985) correlate with the estimated age of the B–$A1$ rotation. A great part of the rotation must have been absorbed by the shortening (about 180 km in the Rhenish massif, according to Oncken et al., 1999) and by the thrust. Nevertheless, a post-Early Permian clockwise rotation by 10–15° of the Ardennes allochtonous terrane relative to the Brabant has been evidenced (Edel and Coulon, 1987; Molina-Garza and Zijderveld, 1996).

5. Conclusions and geodynamic evolution of the Variscides

5.1. Conclusions

The main object of this study was to clear the problem of the C and B directions in the Vosges and northern Massif Central and to fit the timing of acquisition of the magnetizations with respect to the Late Visean tightening phase (340–330 Ma). The results show that the C directions occur in both areas. Pre-tectonic components were found at only two sites, all the remaining C components are syn-tectonic overprints acquired during the Late Visean shortening and uplift. The mean direction 145°/12° is consistent with the mean syn-tectonic C_o direction obtained in the previous study. The B components take the prominent place in the paleomagnetic record. They consist of overprints acquired after the Late Visean tightening and, at least partly, during the Namurian extension. The study confirms that on Paleozoic outcrops, at least 90% of the remanent components yielded after demagnetizations are overprints.

The similar metamorphic, magmatic and tectonic evolutions of the northeastern Massif Central and the southern Vosges in Late Devonian–Carboniferous times (Schneider et al., 1989; Schneider, 1994) and the overlapping paleomagnetic records lead to a common geodynamical evolution of both areas. A common APWP is proposed (Fig. 7). It supports previous geotectonic interpretations (Edel and Wickert, 1991; Edel and Lewandowski, 1993) that considered the C_p–B and the B–$A1$ segments of the APWP in terms of two successive, clockwise, block rotations. In these papers an alternative solution, based on the antipodal C poles and consisting of a succession of a counterclockwise C–B rotation and of a clockwise B–$A1$ rotation, was proposed. According to the timing of the C_o and C_p components deduced from the radiometric ages and from fold tests, this solution seems highly improbable. The clockwise C_p–B, 70° rotation must have been rapid because the overprinting process was interrupted (Edel and Lewandowski, 1993; Edel et al., 1998; in preparation). This is confirmed by the small difference between the age at 330–328 Ma of the youngest pre-rotation C_p components and the age at 327–325 Ma of the collision of the Saxothuringian zone with the Rhenohercynian foreland (Ahrendt et al., 1983). On the contrary, the B–$A1$, 45° rotation lasted for a longer time, as quite a continuous overprinting and several field reversals were recorded from the mean B to the mean $A1$ poles (Fig. 7b). The bulk of the rotation was achieved at the beginning of the Kiaman reversal. The eastern Variscides, Bohemian Massif, Sudetes and even the Upper Silesian platform and Holy Cross Mts (Lewandowski, 1993, 1994) have likely undergone the same motions. In the latter cases the motions were associated with a significant dextral strike-slip motion (Lewandowski, 1994).

These paleomagnetic results imply a much more mobilistic evolution of the Variscan belt than it is generally admitted, and an unconventional Early Carboniferous paleogeography (Fig. 9). The mean

C_o pole and the poles at 340–335 Ma of Laurentia (Torsvik et al., 1996) locate the central Variscides close to the northeastern corner of Baltica, nearly in the continuity of the southern Oural. The C_o–C_p segment of poles involves a fast, latitudinal drift to the south by about 2200 km along the eastern margin of Baltica during the Late Visean (340–328 Ma).

5.2. The geodynamic evolution of the Variscides

In light of the paleodeclinations and paleolatitudes derived from the paleomagnetic results and of the geological constraints, the following history is proposed (Fig. 9).

During the time range 345–340 Ma, in the context of orogen-parallel extension and wrenching, the Variscides are intruded by huge volumes of magmatic rocks. In the Saxothuringian zone, the arc magmatism is related to a northward subducting Rhenohercynian oceanic basin.

From 340 up to 330–328 Ma, the internal Variscides undergo a probably large N–S drift along the eastern margin of Baltica and are affected by wrenching, rapid uplift, thrusting and syn- to post-tectonic magmatism. The C_o and C_p components are recorded, respectively, in the early and late stages of the tectonic phase (Sudetic).

Around 328 Ma, the drift motion turns to a rapid clockwise rotation by about 70° (C_p–B) that closes the Lizard–Giessen oceanic basin. The Eulerian pole of the rotation is located at the southeastern corner of the Baltica margin.

After the collision, large-scale magnetic overprinting (B components) in relation with low-grade metamorphism and extension tectonics affects the internal Variscides, while in the north, the Rhenohercynian sedimentary cover begins to be shortened.

In the Late Namurian–Westphalian, the convergence of Gonwana and Laurussia gives rise to a second clockwise rotation of the whole Variscides east of the Iberian–Armorican arc by 45°. The Rhenohercynian cover is thrusted on the London–Brabant block. The B and $A1$ poles locate the Eulerian pole of the rotation on the southern margin of Baltica.

After acquisition of the primary and secondary $A1$ magnetizations, the clockwise rotation of the Variscides continues, but from now on, together with whole Laurussia. The global motion lasts up to the end of the Alleghanian convergence in the Middle Permian (270 Ma) (Edel and Schneider, 1995).

Acknowledgements

I wish to thank J.P. Braun for his efficient help in the field. I acknowledge J.P. Burg, M. Mattei, T. Torsvik, R. Van der Voo and my collegues R. Montigny and M. Westphal for critical reading of the manuscript and constructive comments. I particularly appreciate the stimulating remarks and suggestions of M. Lewandowski. This is a contribution of UMR 7516 Institut de Physique du Globe of C.N.R.S and Université Louis Pasteur of Strasbourg.

References

Ahrendt, H., Clauer, N., Hunziker, J.C., Weber, K., 1983. Migration of folding and metamorphism in the Rheinische Schiefergebirge deduced from K–Ar and Rb–Sr age determinations. In: Martin, H., Eder, F.W. (Eds.), Intracontinental Fold Belts. Springer, Berlin, pp. 324–338.

Bachtadse, W., Heller, F., Kröner, A., 1983. Paleomagnetic investigations in the Hercynian Mountain belt of Central Europe. Tectonophysics 91, 285–299.

Boutin, R., Montigny, R., Thuizat, R., 1995. Chronologie K–Ar et ^{39}Ar–^{40}Ar du métamorphisme et du magmatisme des Vosges, comparaison avec les massifs varisques avoisinants. Géol. France 1, 3–25.

Coulon, M., Fourquin, C., Paicheler, J.C., 1979. Contribution du tectogène varisque dans les Vosges méridionales, III-Le Culm entre Bourbach-le-Haut et le Molkenrain. Sci. Géol. Bull. 32/3, 117–129.

Dallmeyer, R.D., Neubauer, F., Höck, V., 1992. Chronology of late Paleozoic tectonothermal activity in the southeastern Bohemian Massif, Austria (Moldanubian and Moravo-Silesian zones): $^{40}Ar/^{39}Ar$ mineral age control. Tectonophysics 210, 135–153.

Edel, J.B., 1987a. Paleomagnetic evolution of the Central Massif (France) during the Carboniferous. Earth Planet. Sci. Lett. 82, 180–197.

Edel, J.B., 1987b. Paleopositions of the western Europe Hercynides during the late Carboniferous deduced from paleomagnetic data: consequences for stable Europe. Tectonophysics 139, 31–42.

Edel, J.B., 1999. Magnetic overprints: a tool to date hydrothermal activity. The example of the Limousin (Massif-Central, France). EUG X meeting, March 1999, Strasbourg, vol. 11, p. 478 (Terra Abstracts).

Edel, J.B., Coulon, M., 1987. A paleomagnetic cross-section through the Ardenne and the Brabant Massif (France–Belgium). J. Geophys. 61, 21–27.

Edel, J.B., Duringer, P., 1997. Paleomagnetic constrains on the Apparent Polar Wander Path of the European Plate in Late

Triassic–Early Jurassic times — new data from NW France and SW Germany. Geophys. J. Int. 128, 331–344.

Edel, J.B., Wickert, F., 1991. Paleoposition of the Saxothuringian (Northern Vosges, Pfalz, Odenwald, Spessart) in Variscan times: paleomagnetic investigation. Earth Planet. Sci. Lett. 103, 10–26.

Edel, J.B., Lewandowski, M., 1993. Polyphased palaeomagnetic overprinting in the central part of the Variscan belt, with emphasis on the southern Vosges Devonian–Dinantian basin. Geotectonic interpretation. New Trends in Geomagnetism, IIIrd biennial Meeting, Smolenice, Slowakia, June 1992, Geol. Carpathica 44/6, 381–398.

Edel, J.B., Schneider, J.L., 1995. The Late Carboniferous to Early Triassic evolution of Variscan Europe in the light of magnetic overprints in Early Permian rhyolites from northern Vosges (France) and central Black Forest (Germany). Geophys. J. Int. 122, 858–876.

Edel, J.B., Weber, K., 1995. The Cadomian terranes, wrench-faulting and thrusting in the Central Europe Variscides-geophysical and geological evidence. Geol. Rundsch. 84, 412–432.

Edel, J.B., Lacaze, M., Westphal, M., 1981. Paleomagnetism in the north-eastern Central Massif (France), evidence for Carboniferous rotations of the Hercynian orogenic belt. Earth Planet. Sci. Lett. 55, 48–52.

Edel, J.B., Humbert, C., Honnorez, J., 1998. Asynchronous magnetic overprinting in the Visean volcanic layers of the southern Vosges (France) as a tracer of the APWP of Variscan Europe from the Visean to the Jurassic. In EGS XXIII Ass., Nice, April 1998, Abstr.

Edel, J.B., Montigny, R., Royer, J.Y., Thuizat, R., Trolard, F., 1986. Paleomagnetic investigations and K–Ar dating on the Variscan plutonic massif of the Champ du Feu and its volcanic sedimentary environment, northern Vosges, France. Tectonophysics 122, 165–185.

Edel, J.B., Aïfa, T., Jelenska, M., Kodziako-Hofmokl, M., Zelazniewicz, A., 1997. Réaimantations des formations paléozoïques des Sudètes polonaises et courbe de dérive des pôles géomagnétiques d'Europe du Carbonifère moyen au Jurassique moyen. C. R. Acad. Sci. Paris, Earth Planet. Sci. 325, 479–486.

Hess, J.C., Lippolt, H.J., Kober, B., 1995. The age of the Kagenfels granite (northern Vosges) and its bearing on the intrusion scheme of late Variscan granitoids. Geol. Rundsch. 84, 568–577.

Jelenska, M., Kadzialko-Hofmokl, M., Edel, J.B., Jamrozik, L., Petersen, N., Soffel, H., 1995. Palaeomagnetic investigations of the Palaeozoic circum-Sowie Gory Mountains ophiolitic belt in the Sudetes, Poland. Geophys. J. Int. 122, 658–674.

Kent, J.T., Briden, J.C., Mardia, K.W., 1983. Linear and Planar structure in ordered multivariate data as applied to progressive demagnetization of palaeomagnetic remanence. Geophys. J. R. Astron. Soc. 75, 593–621.

Kirschvink, J.L., 1980. The least square line and plane and the analysis of paleomagnetic data. Geophys. J. R. Astron. Soc. 62, 699–718.

Kreuzer, H., Harre, W., 1975. K/Ar Altersbestimmungen an Hornblenden und Biotiten des Kristallinen Odenwaldes. Aufschluss Sdb. 27, 71–77.

Krohe, A., 1992. Structural evolution of intermediate-crustal rocks in a strike-slip and extensionnal setting (Variscan Odenwald, SW Germany): differential upward transport of metamorphic complexes and changing deformation mechanisms. Tectonophysics 205, 357–386.

Krs, M., Pruner, P., 1995. Paleomagnetism and paleogeography of the Variscan formations of the Bohemian Massif. J. Czech. Geol. Soc. 40, 3–45.

Lewandowski, M., 1993. Paleomagnetism of the Paleozoic rocks of the Holy Cross Mts (Central Poland) and the origin of the Variscan orogen. Publ. Inst. Geophys. Pol. Acad. Sci. A-23 (265), 1–85.

Lewandowski, M., 1994. Palaeomagnetic constraints for Variscan mobilism of the Upper Silesian and Malopolska Massifs, southern Poland. Geol. Q. 38, 211–230.

Lowrie, W., 1990. Identification of ferromagnetic minerals in a rock by coercitivity and unblocking temperature properties. Geophys. Res. Lett. 17/2, 159–162.

Matte, P., 1991. Accretionary history and crustal evolution of the Variscan belt in Western Europe. Tectonophysics 196, 309–337.

Matte, P., Hirn, A., 1988. Seismic signature and tectonic cross section of the Variscan crust in western France. Tectonics 7/2, 141–155.

Matte, P., Maluski, H., Rajlich, P., Franke, W., 1990. Terrane boundaries in the Bohemeian Massif: result of a large-scale Variscan shearing. Tectonophysics 177, 151–170.

Mauritsch, H.J., Pruner, P., 1990. Paleomagnetic investigations in the central part of the Bohemian Massif. In: Minarikova, D., Lobitzer, H. (Eds.), Thirty years of Geological cooperation between Austria and Czechoslovakia. Federal Geology Survey of Vienna, Geology Survey of Prague, pp. 56–64.

McElhinny, M.W., 1964. Statistical significance of the fold test in palaeomagnetism. Geophys. J. R. Astron. Soc. 8, 338–340.

McFadden, P.L., 1990. A new fold test for paleomagnetic studies. Geophys. J. Int. 103, 163–169.

Molina-Garza, R.S., Zijderveld, J.D.A., 1996. Paleomagnetism of Paleozoic strata, Brabant and Ardennes Massifs, Belgium: implications of prefolding and postfolding Late Carboniferous secondary magnetizations for European apparent polar wander. J. Geophys. Res. 101 (B7), 15799–15818.

Nawrocki, J., 1993. The Devonian-Carboniferous platform paleomagnetic directions from the Silesian-Crackow area and their importance for Variscan paleotectonic reconstructions. Geol. Q. 37, 397–430.

Oncken, O., 1998. Evidence for precollisional subduction erosion in ancient collisional belts: the case of the Mid-European Variscides. Geology 26, 1075–1078.

Oncken, O., von Winterfeld, C., Dittmar, U., 1999. Accretion of a passive rifted margin: the Late Paleozoic Rhenohercynian fold and thrust belt (Middle European Variscides). Tectonics 18, 75–91.

Petrini, K., Burg, J.P., 1998. Relationship between deformation, plutonism and regional metamorphism in the Markstein area (southern Vosges). Géol. France 2, 13–23.

Piqué, A., Huon, S., Clauer, N., 1984. La schistosité hercynienne et le métamorphisme associé dans la vallée de la Meuse, entre Charleville-Mézières et Namur (Ardennes franco-belges). Bull. Soc. Belg. Géol. 93, 55–70.

Reisinger, J., Edel, J.B., Mauritsch, H.J., 1994. Late Carboniferous-Late Permian Paleomagnetic overprinting of Carboniferous granitoids in southern Bohemian Massif (Austria). Phys. Earth Planet. Int. 85, 53–65.

Roques, M., Vachette, M., Vialette, Y., 1971. Géochronologie du socle du Massif Central. In Syposium J. Jung, Géologie, géomorphologie et structure profonde du Massif Central Français, pp. 269–289.

Schaltegger, U., 2000. Geochronology of the Southern Black Forest batholith (Central Variscan Belt): timing of exhumation and granite emplacement. Int. Earth Sci. (Geol. Rundschau) 88, 814–828.

Schaltegger, U., Schneider, J.L., Maurin, J.C., Corfu, F., 1996. Precise U–Pb chronometry of 345–340 Ma old magmatism related to syn-convergence extension in the southern Vosges (Central Variscan belt). Earth Planet. Sci. Lett. 144, 403–419.

Schaltegger, U., Maurin, J.C., Schulmann, K., Fanning, M., 1997. U–Pb single-grain and SHRIMP zircon dating of granulites and migmatites of the Vosges mountains: multiple melting during rapid exhumation. Abstr. Terra Nova 9, 112 (EUG meeting 1997: Orogenic processes).

Schneider, J.L., 1990. Enregistrement de la dynamique varisque dans les bassins volcano-sédimentaires dévono-dinantiens exemple des Vosges du Sud (zone moldanubienne). Thèse, Strasbourg I, 222 pp.

Schneider, J.L., 1994. Le bassin volcano-sédimentaire dévono-dinantien des Vosges du Sud: un temoin de l'évolution de la chaîne varisque d'Europe. Bull. Soc. Hist. Nat. Colmar 62, 49–86.

Schneider, J.L., Maas, R., Gall, J.C., Duringer, P., 1989. L'événement intraviséen dans la zone moldanubienne de la chaîne varisque d'Europe: les données des formations volcano-sédimentaires dévono-dinantiennes du Massif Central Français, des Vosges du Sud (France) et de la Forêt Noire (R.F.A.). Geol. Rundsch. 78/2, 555–570.

Schulmann, K., Ledru, P., Autran, A., Edel, J.B., 1997. Thermo-mechanical evolution of Variscan collision in western and Central Europe: petrological, structural and geophysical constrains. Abstr. Terra Nova 9, 112 (EUG meeting 1997: Orogenic processes).

Spaeth, G., Fiedlitz, W., Frank, B., 1985. Caledonian deformation and very low-grade metamorphism in the north-eastern part of the Stavelot-Venn Massif, Neues Jahrb. Geol. Paläontol: Abh 171, 297–310.

Stipska, P., Schulmann, K., 1995. Inverted metamorphic zonation in a basement-derived nappe sequence, easten margin of the Bohemian Massif. Geol. J. 30, 385–413.

Tait, J., Bachtadse, V., Soffel, H., 1996. Eastern Variscan fold belt: paleomagnetic evidence for oroclinal bending. Geology 24, 871–874.

Torsvik, T.H., Smethurst, M.A., Meert, J.G., Van der Voo, R., McKerrow, W.S., Brasier, M.D., Sturt, B.A., Walderhaug, H.J., 1996. Continental break-up and collision in the Neoproterozoic and Paleozoic — a tale of Baltica and Laurentia. Earth Sci. Rev. 40, 229–258.

Van der Voo, R., 1990. Phanerozoic paleomagnetic poles from Europe and North America and comparison with continental reconstructions. Rev. Geophys. 28/2, 167–206.

Vennat, G., 1985. Le complexe volcano-plutonique viséen supérieur de Gannat-les-Ancizes (Allier, Puy de Dôme). Etudes pétrographiques, structurales, géochimiques et dynamiques. Géol. France 4, 341–364.

Vialette, Y., 1962. Contribution à l'étude géochronologique par la méthode au strontium des principaux massifs de granite et de migmatites du Massif Central français. Ann. Fac. Sci. Univ. Clermont, vol. 6, 88 pp.

Watson, G.S., Enkin, R.J., 1993. The fold test in paleomagnetism as a parameter estimation problem. Geophys. Res. Lett. 20, 2135–2137.

Weber, K., Behr, H.J., 1983. Geodynamic interpretation of the Mid-European Variscides. In: Martin, H., Eder, F.W. (Eds.), Intracontinental fold belts. Springer, Berlin, pp. 427–469.